AMARTYA RANJAN

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SUMMARY

Final-year B.Tech student in Computer Science (AI & Robotics) with production experience building AI/ML systems, LLM-powered applications, and multimodal machine learning pipelines. Experienced in PyTorch, deep learning, NLP, reinforcement learning, generative AI, RAG, and agentic systems, with hands-on experience in GPU programming, kernel optimization, model inference, and ML system development. Experienced in building and deploying production AI pipelines and research-oriented ML systems.

EDUCATION

Vellore Institute of Technology, Chennai | 2023 – 2027
B.Tech – Computer Science (AI & Robotics) | **CGPA: 8.73 / 10**

Delhi Public School, Varanasi | 2023
Higher Secondary Education (XII) – CBSE | **95%**

EXPERIENCE

AI Developer Intern | Dextora | Jan 2026 – Aug 2026

- Built and deployed a **4-phase self-healing RAG pipeline** for a client-facing production system, integrating automated ingestion, preprocessing, retrieval, evaluation, and feedback loops; improved answer relevance by **8 points over a naive RAG baseline**.
- Developed production ML data-processing and evaluation pipelines using Python, Pandas, NumPy, text preprocessing, chunking, feature extraction, and automated evaluation.
- Architected an **agent orchestration system** using structured task representations and specialized sub-agents to generate and execute dynamic dispatch logic.
- Built an end-to-end **AI video generation pipeline** integrating web crawling, data extraction, FFMPEG-based media processing, and automated scene assembly.
- Worked with **LangChain and LangGraph** to implement agentic workflows, state management, tool execution, and multi-step reasoning loops.

Research Intern | CeHMA – VIT Chennai | 2025

- Designed and trained **multimodal ML pipelines** combining clinical records, medical imaging, genetic markers, and demographic data for Parkinson's disease progression prediction.
- Performed data preprocessing, feature engineering, heterogeneous data integration, and predictive model development.
- Identified important non-motor biomarkers through data-driven analysis and contributed to an interpretable clinical decision-support framework.

PROJECTS

Self-Healing Hybrid RAG Engine | Python, RAG, LLM Evaluation, LangGraph

- Designed a **closed-loop RAG architecture** that detects weak retrieval results, automatically triggers corrective processing, and uses feedback-driven components to improve retrieval quality.
- Improved answer relevance by **8 points over a naive RAG baseline** through automated retrieval evaluation and corrective processing.

High-Efficiency Deep Handwriting Transcriber | PyTorch, Bi-Axial LSTM, CTC Loss

- Designed a **10-layer Bi-Axial LSTM** architecture for full-text handwriting transcription, achieving **11.4% Character Error Rate (CER)**.
- Applied spatial parallelisation techniques to improve computational efficiency, achieving a **19× speedup over MDLSTM** while maintaining accuracy.

Prodromal Parkinson's Prediction | PyTorch, Multimodal Learning, Machine Learning

- Built a multimodal prediction pipeline integrating **four heterogeneous data sources**, feature extraction, preprocessing, and predictive modelling for Parkinson's disease progression.
- Developed data integration and feature-engineering workflows for combining heterogeneous clinical and biomedical signals.

AI Healthcare Policy Simulator | Python, Reinforcement Learning, Simulation

- Processed and simulated **49,736 AI household agents** modeled on 2011 Indian census data to evaluate healthcare policy interventions at national scale.
- Applied reinforcement learning and decision modelling under India-specific socioeconomic constraints.

Diffusion Model from Scratch | PyTorch, Diffusion Models, VAE, U-Net

- Implemented a **diffusion-model pipeline from scratch** using a VAE and U-Net architecture for synthetic data generation.
- Implemented the core training and generation pipeline without relying on a high-level diffusion framework.

TECHNICAL SKILLS

Programming: Python, C++, C, Java, CUDA Python
Machine Learning: PyTorch, Scikit-learn, Deep Learning, NLP, Multimodal Learning, Reinforcement Learning, Time-Series Modeling, Feature Engineering
Generative AI & LLMs: RAG, LLM Evaluation, Agentic AI, ReAct Agents, LangChain, LangGraph, vLLM, LLM Inference
ML Systems: Model Training, Model Evaluation, ML Pipelines, Model Inference, Distributed Processing
GPU & Performance Engineering: CUDA, Kernel Programming, Kernel Fusion, Numba, JAX, GPU Optimization
Data & ML Infrastructure: Pandas, NumPy, SQL, PostgreSQL, Apache Spark, Apache Kafka, Apache Airflow, ETL/ELT
Engineering: Docker, Linux, Git, GitHub, FFMPEG, Jupyter

ACHIEVEMENTS

- **National-level Hackathon Finalist** – Top 15 teams for a SEBI scam-detection machine learning model.

RESEARCH INTERESTS

Deep Learning · Generative AI · LLMs · Multimodal Learning · Reinforcement Learning · ML Systems · GPU Computing · Efficient Inference · Agentic AI